DEGREE

Degree, also known as *degree centrality*, is the total number of links that are connected to a node. In our example data, Mo has a degree of 6, because he's the source or target of 6 links. Degree is a rough measure of the importance of a node in a network, because you assume that people or things with more connections have more power or influence in a network. Weighted degree is used to refer to the total value of the connections to a node, which would give Mo a value of 18. Further, you can differentiate degree into *in degree* and *out degree*, which are used to distinguish between incoming and outgoing links, and which for Mo's case would be 4 and 2, respectively.

Every time you start the `force()` layout, D3 computes the total number of links per node, and updates that node's weight attribute to reflect that. We'll use that to affect the way the force layout runs. For now, let's add a button that resizes the nodes based on their `weight` attribute:

```
d3.select("#controls").append("button")
      .on("click", sizeByDegree).html("Degree Size");

function sizeByDegree() {
   force.stop();
   d3.selectAll("circle")
      .attr("r", function(d) {return d.weight * 2;});
};
```

Figure 6.11 shows the value of the degree centrality measure. Although you can see and easily count the connections and nodes in this small network, being able to spot at a glance the most and least connected nodes is extremely valuable. Notice that we're counting links in both directions, so that even though Tully is connected to

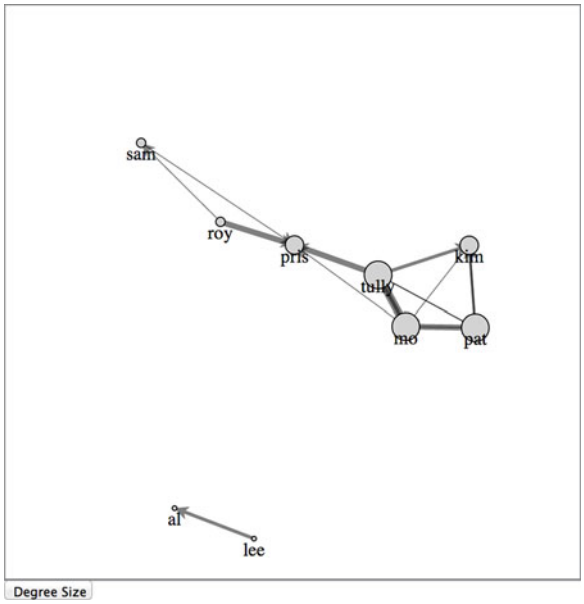


Figure 6.11 Sizing nodes by weight indicates the number of total connections for each node by setting the radius of the circle equal to the weight times 2.